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The Anatomy of the Strait of Hormuz Oil Shock

1. Introduction

Every oil shock has its own special features, and the current one is no different. Until before the start of the US-Israel war on Iran, most observers viewed a long-term disruption of the Strait of Hormuz (SOH) as a low-probability high-impact event confined to war game exercises. While during the 1984-1988 Iran-Iraq Tanker War, attacks on vessels did occur leading to an increase in insurance premiums and freight rates and a decrease in the number of vessels passing through the SOH, oil flows did not stop. This view of uninterrupted oil flows underscored many fundamental beliefs about the functioning of the oil market including the resilience of oil markets and its physical and financial infrastructure to various types of shocks as well as the interconnectedness of global oil markets via a wide network of trade logistics, transport routes, infrastructure and market players which ensured an efficient and secure flow of oil. The disruption of oil flows through the SOH, which comes on the heels of the rerouting of oil flows following the Russia-Ukraine war, and the increased uncertainty around the post-war framework that would govern transit through the SOH are key distinguishing features of the current shock and are causing a major revision of key aspects of the oil market, energy policies, security arrangements, and market players' behaviours and strategies.

Another important feature of the current crisis has been the size of the supply shock compared to the existing buffers in the oil system. These buffers, mainly in the form of commercial and strategic stocks, are volumetric in nature and help offset production losses, easing the pressure on the oil price to adjust sharply to restore market equilibrium. The current war has resulted in large production losses. Total oil liquids output from the Gulf is estimated to have fallen by close to 12.6 million b/d (mb/d) in March relative to pre-war February 2026 levels, with the disruption expected to increase to 13.2 mb/d in April.[1] On this basis, the cumulative output loss since the start of the war is projected to reach around 790 million barrels by the end of April. At the early stages, the market's focus was on quantifying output losses. As the disruption persisted, this focus broadened to estimating the time needed to restore production to pre-war levels and the cost of repairs, and whether some of the Gulf producers could permanently lose part of their productive capacity. The existing buffers in the oil system are insufficient to absorb a shock of this magnitude in volumetric terms, particularly because spare capacity, one of the market's most effective adjustment mechanisms, is unavailable as it is located in the same region as the source of the disruption. The remaining buffers in the system, such as commercial and strategic petroleum reserves, are small compared to the size of the supply shock. Thus, the SOH disruption does not only represent a flow and production shock; it has also eroded the effectiveness of a key mitigation and an important adjustment mechanism available for oil markets.

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[1] OIES. OIES Oil Monthly Issue 53, 23 April 2026.

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Another feature of the current crisis is that it is not only a shock to crude oil, but also to supplies of refined products, with significant implications for end users in the form of higher prices and reduced availability of refined products particularly in Asia, with the shock spreading to other regions such as Africa and Europe. This in part reflects the increasing importance of the Gulf producers, particularly those in the Gulf Cooperation Council (GCC), in refining and downstream. It also highlights the interdependence between Asia, the world's fastest growing region, and the Gulf. Crude and products from the Gulf represent the baseload for Asian refineries and petrochemical industries, while Asia is a key export market for the Gulf petroleum sector.

Over the years, the oil market has become more financialized and futurized with a very large derivatives ecosystem and more varied pricing system particularly in Asia.[2] Consequently, the shock in oil markets' physical layers propagated to its financial layers, impacting flows into oil derivatives, producers' and refineries' hedging positions, the financial position of physical and financial traders, price volatility, and the links between prices in physical and futures markets.[3,4] The impacts of the current shock extended beyond futures to over-the-counter (OTC) swaps, reducing liquidity in key markets. This impaired the price discovery function of key benchmarks in the Gulf, and contributed to increased volatility and erratic price movements, and caused massive losses as some players lost the physical leg of their hedges.

The supply shock is also occurring against a highly geopolitically fragmented world in which oil has been widely used as a tool to influence foreign policy and trade outcomes. Some of the world's top oil producers, Iran, Russia, and Venezuela have been subject to sanctions, embargoes, blockades and price caps resulting in highly segmented and a less transparent market, more complex trade logistics, and increased importance of the dark fleet in transporting and storing crude, intensifying the challenge of tracking trade flows. Energy infrastructure, such as refineries, tankers and ports, has also been subject to regular attacks. Trade routes were disrupted causing long and costly diversions. And the US has imposed tariffs on importers of sanctioned crude as part of its trade and foreign policy. The current war will only deepen the existing geopolitical fractures impacting oil market volatility, the structure of oil markets, and trade flows, oil relations, and investment decisions.

Every crisis both shapes and is shaped by the strategies and behaviour of key players. The key player in this war has been the US. In addition to leading the war against Iran, the Trump Administration has created the main headlines that moved oil prices, heightened volatility, and shaped sentiment. In its attempt to put a lid on the oil price, it resorted to multiple tools including influencing the narrative through a heavy flow of market-moving headlines, releasing crude stocks from the SPR and granting waivers on sanctioned Russian and Iranian barrels to provide short-term relief to oil markets. In addition, the importance of WTI in global trade flows and in setting the price of key benchmarks continues to increase, with US exports of crude and products hitting record levels and reaching all over the globe. However, current events also revealed the limits to US energy dominance. The US can't isolate itself from higher oil prices[5], and rather than securing a stable environment which incentivises investment and growth in oil production, it has created a more volatile and an uncertain environment even for US producers. Another key player has been Iran. As part of its broader strategy to impose maximum pain on the global economy, Iran has attacked energy infrastructure in the Gulf and disrupted free passage through the SOH, demonstrating its readiness to weaponize the world's most important oil transit chokepoint and in the process has dismantled the security arrangements that prevailed for decades. There have been some 'accidental' winners, in particular Russia, which has been able to reprice its crude, increase revenues and expand its client base, but has been limited in its ability to increase its exports due to the continued Ukrainian attacks on its physical infrastructure. While China has entered the war in a relatively strong position to deal with the shock thanks to its strategy of accumulating high levels of crude stocks, a longer duration war and/or a prolonged US blockade of the SOH will test China's oil supply security. Asian countries outside China have felt the brunt of the supply shock which

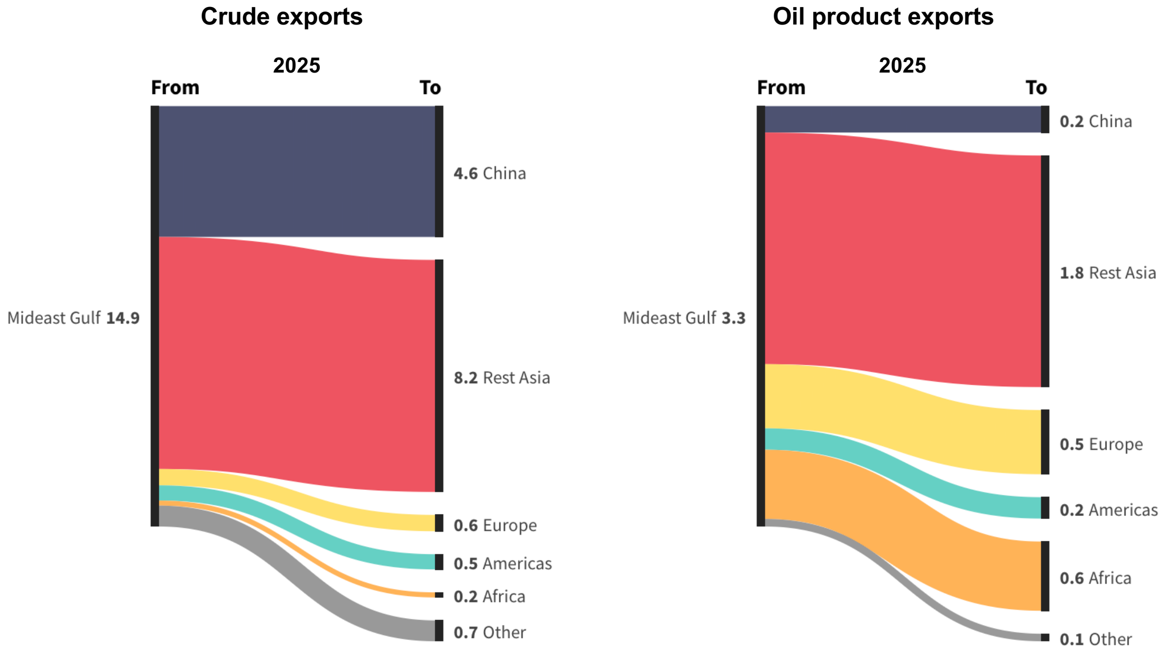
[2] Fattouh, B. and Mehdi, A. 2025.'Dubai, We Have a Problem: Murban and Middle East Crude Pricing', OIES Comment, March.
[3] Bouchouev, I. 2026. 'Energy Quantamentals: The Oil Crisis in the Eyes of a Financial Trader', OIES Comment, April.
[4] Fattouh, B. and Mehdi, A. 2026.'Through the Looking Glass: Oil and the Search for Direction', OIES Comment, March.
[5] Financial Times. 'Impact of Iran war will hurt US even after conflict ends, economists warn', 19 April 2026.

required some severe adjustments in the form refining cuts and demand rationing. Elsewhere, even regions with lower direct dependence on SOH oil flows are not immune to the shock. Europe faces tougher competition for crude and product supplies, and potential cuts in refining runs impacting the availability of key products such as jet fuel. Africa's limited refining capacity renders the region highly vulnerable to refined product tightness, particularly in product-import-dependent economies that may struggle to secure replacement cargoes. Ultimately, the current oil shock has exposed the limits of regional insulation: product tightness and higher benchmark prices are transmitted globally, feeding into inflationary and macroeconomic pressures even in economies with limited direct dependence on SOH flows.

2. An oil shock like no other

A key feature that sets the current shock apart is its sheer magnitude. The current episode constitutes the most severe flows and supply disruption in the history of oil markets. The paramount importance of the SOH lies in the fact that it remains the main principal artery through which Gulf crude and products reach global markets. Based on 2025 annual averages, the Strait carried about 25% or 18.2 mb/d of global oil liquids exports, including 14.9 mb/d of crude and condensates and 3.3 mb/d of refined products (**Figure 1**). Asia was the main destination, receiving roughly 80% of total flows, and within Asia, China accounted for nearly one-third of total exports at close to 5 mb/d, followed by India, South Korea and Japan, each at about 2 mb/d (or around 12%). Europe received around 6% of the total, or about 1.1 mb/d, and the Americas and Africa each accounted for less than 5%.

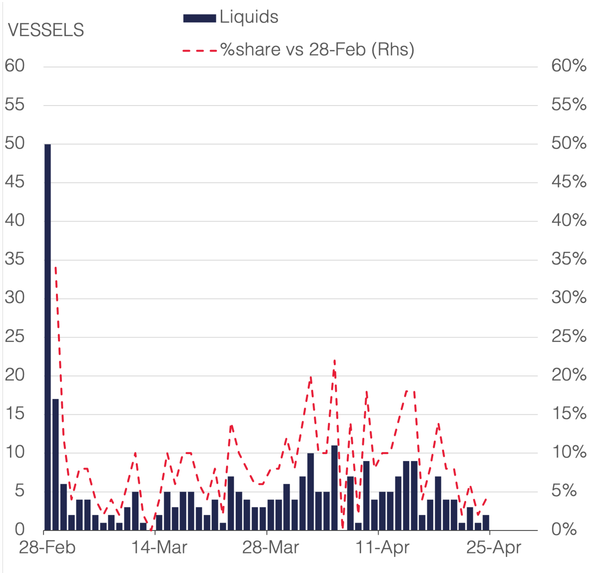
Fig. 1: Middle East Gulf crude and oil product exports via the Strait of Hormuz, in mb/d



Notes: Includes crude/co, clean products and dirty petroleum products.
Source: Kpler, OIES

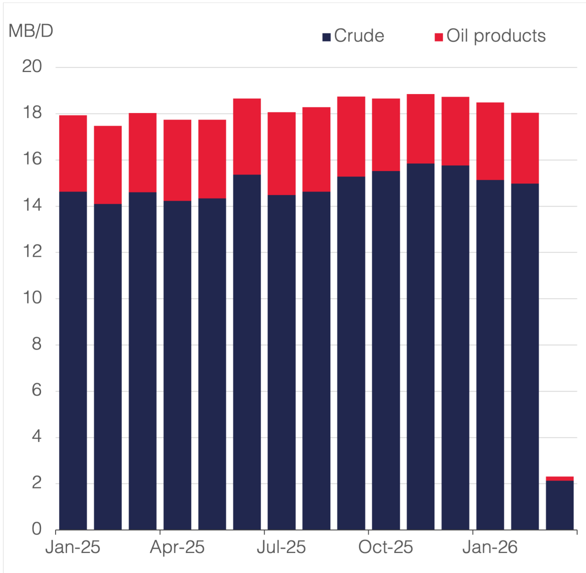
In 2025, crude and NGLs output from Gulf countries–namely Saudi Arabia, Iraq, Iran, Kuwait, the UAE, Qatar and Bahrain–accounted for roughly 28% of global supply (30 mb/d). Bypass options exist, but they are limited and concentrated in only two Gulf producers, Saudi Arabia and the UAE. The 7 mb/d East-West pipeline, which had a run rate of 2.6 mb/d in 2025, has been a key tool for redirecting volumes from Ras Tanura to Yanbu on the Red Sea. It is estimated that loading capacity from the Yanbu terminal stands at 4.8 mb/d of crude and 900 thousand barrels per day (kb/d) of products. The UAE, whose pre-war crude loadings via the Strait averaged near 2.3 mb/d, can redirect flows via the ADCOP pipeline to the Fujairah terminal on the Gulf of Oman that has nameplate capacity of 1.7 mb/d. The remaining Gulf producers are overwhelmingly dependent on the SOH for their oil exports, with only limited exceptions. Iraq has some restricted rerouting capacity via the Kirkuk-Ceyhan pipeline. Nameplate capacity is around 1.6 mb/d, but effective capacity is between 400 kb/d and 500 kb/d due to outages, damages and downstream constraints, which is nowhere near the scale required to restore the more than 3 mb/d Iraqi southern exports lost through Hormuz. Iran can bypass the SOH to the Gulf of Oman via the Goreh-Jask pipeline, but again with limited effective capacity of 300 kb/d.

Fig. 2: Strait of Hormuz crossings



Notes: Both directions, as of 24 April 2026.
Source: Kpler, OIES

Fig 3: Middle East Gulf crude and oil product exports via the Strait of Hormuz



Source: Kpler, OIES

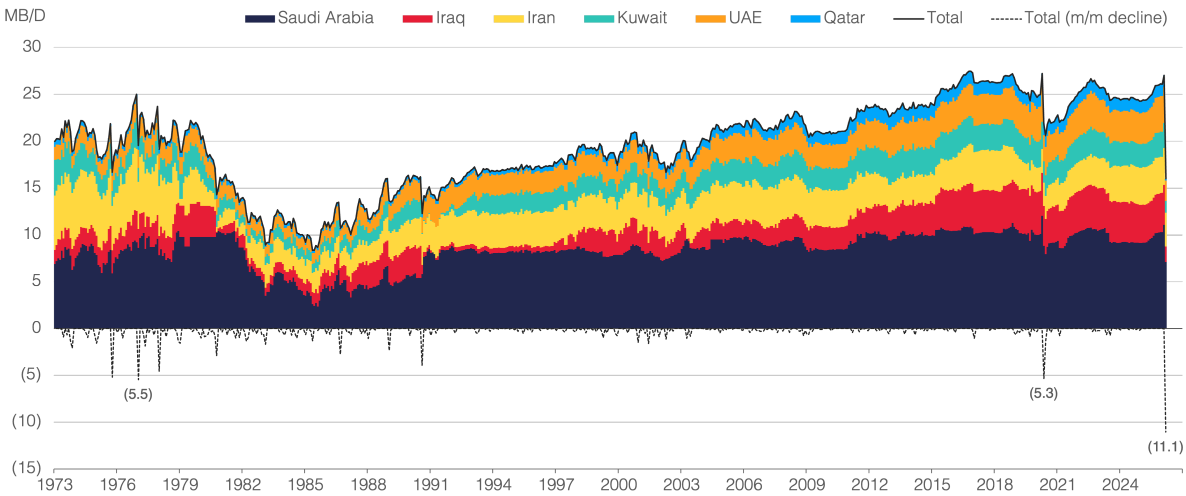
At the time of writing, the SOH has remained effectively shut for 55 days, from 1 March to 24 April, with observed average crossings in both directions down by more than 90% relative to pre-war levels (**Figure 2**). According to Kpler data, observed daily liquid-cargo crossings in and out of the Strait averaged about four per day, down from roughly 50 on 28 February 2026, immediately before the war. Total crude and products exports in March averaged 2.3 mb/d, almost 90% below the pre-war February level of 18 mb/d (**Figure 3**), of which 75% or 1.7 mb/d originated from Iran. On 13 April 2026, following a failed first round of US-Iran negotiations in Pakistan, the US began a blockade of Iranian port traffic designed to halt remaining Iranian exports.[6]

2.1. From a flow shock to a production shock

The disruption of flows through the SOH which remains ongoing, alongside attacks on critical oil infrastructure in the Gulf, quickly transformed the initial flow shock into the largest production shut in at the wellhead. Crude and condensate output from key Gulf producers[7] declined by 11.1 mb/d month-on-month (m/m) in March, falling to 16 mb/d from 27 mb/d in February (see **Figure 4**). This represents the largest m/m supply disruption in the history of the global oil market, and is larger than the voluntary monthly decline implemented in May 2020 in response to COVID.

[6] Bloomberg. 'US Says It Will Intercept, Divert or Capture Ships Leaving Iran', 13 April 2026.
[7] Includes Saudi Arabia, Iraq, Iran, Kuwait, the UAE, Qatar, Oman and Bahrain.

Fig. 4: Historical Middle East Gulf crude oil production



Notes: Data shown from January 1973 to March 2026. Includes crude/co.
Source: WTRG Economics, IEA, OPEC, OIES

The biggest impacts are observed in countries without bypass capacity, namely Qatar (-85% m/m), Iraq (-61% m/m) and Kuwait (-53% m/m), followed by the UAE (-45% m/m) and Saudi Arabia (-36% m/m) (**Table 1**). Iranian output, whose exports were affected far less m/m compared to the rest Gulf producers, declined by only 6% m/m, or 180 kb/d, and held near 3 mb/d but this decline is expected to accelerate following the US blockade.

Table 1: Middle East Gulf oil production losses, in mb/d

	Saudi Arabia	Iraq	UAE	Kuwait	Iran	Qatar	Total
Crude oil production - Feb 26	**10.88**	**4.19**	**3.42**	**2.58**	**3.24**	**1.25**	**25.56**
Target production	10.10	4.27	3.41	2.58	n/a	n/a	
Crude production - Mar 26	**6.96**	**1.62**	**1.89**	**1.21**	**3.06**	**0.18**	**14.93**
Change versus Feb-26 (mb/d)	-3.92	-2.56	-1.53	-1.37	-0.18	-1.07	-10.63
Change versus Feb-26 (%)	-36%	-61%	-45%	-53%	-6%	-85%	-42%
Crude production - Apr 26 F	**7.46**	**1.39**	**1.67**	**0.77**	**2.82**	**0.10**	**14.20**
Change versus Feb-26 (mb/d)	-3.42	-2.80	-1.75	-1.82	-0.42	-1.15	-11.36
Change versus Feb-26 (%)	-31%	-67%	-51%	-70%	-13%	-92%	-44%

Notes: Includes crude oil only. Qatar includes crude/co.
Source: OPEC Secondary sources, OIES forecast

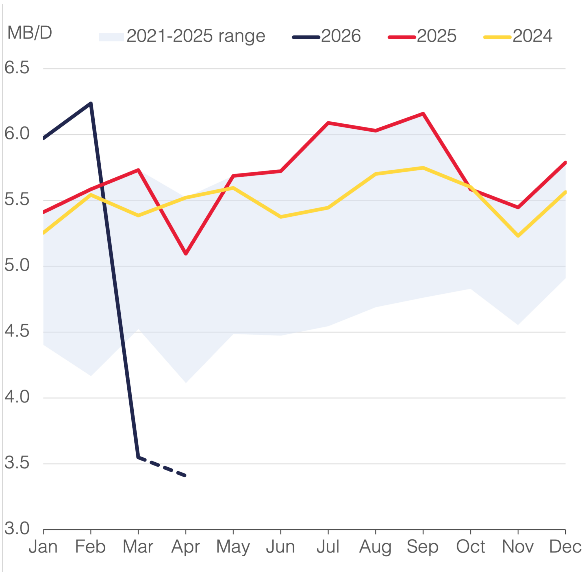
Absent a material breakthrough in US-Iran negotiations and with the US blockade set to tighten further the remaining less than 10% of flows that have so far managed to transit through the Strait, we expect these output losses to deepen further in April, reaching 11.4 mb/d relative to pre-war February levels, and close to 13 mb/d on a total liquids basis, including NGLs (see **Table 1**).[8]

[8] OIES. OIES Oil Monthly Issue 53, 23 April 2026.

2.2. Products supply crunch adds to the severity of the shock

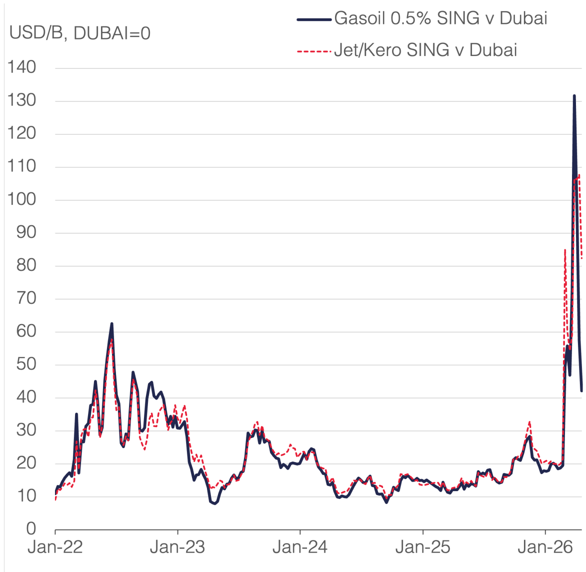
In addition to a crude supply crunch, markets are also facing a refined products supply shock. Oil product flows through the SOH have been heavily disrupted, falling m/m from 3 mb/d in February to 180 kb/d in March, with Asia bearing the brunt of the decline. GCC refinery runs also declined sharply between February and March, falling by 43% m/m from 6.2 mb/d to 3.5 mb/d (**Figure 5**). Multiple refineries in the GCC have been subject to Iranian attacks with varying degree of damage including the Sitra Refinery in Bahrain, the Mina Al Ahmadi and Mina Abdullah refineries in Kuwait, the Ruwais refinery in the UAE, and the SATORP and Ras Tanura refineries in Saudi Arabia. Since the start of the war, an estimated 3.5 mb/d of Gulf capacity has been shut-in either due to physical damages, precautionary closure, and logistical constraints, including storage saturation constraints.

Fig. 5: GCC refinery runs



Notes: April 2026 is based on Kpler predictive.
Source: Kpler, OIES

Fig 6: Asian product cracks



Notes: Data shown are weekly, as of week-ending 24 April 2026. Source: Argus, OIES

Outside the GCC, Asian refineries have been scrambling for barrels, particularly medium sour barrels which form their main baseload. Reduced crude availability and higher crude cost prompted some refineries to cut runs.[9] Asian refineries face higher costs as they also have to compete to secure medium sour crudes from more distant locations in Europe or Latin America with high freight rates exacerbating these costs, though the US waiver on sanctioned Russian and Iranian crude provided some relief. Replacing medium sour crudes with lighter crude also means less efficient operations and less availability of vacuum gasoil (VGO) for hydrocrackers needed for distillate production. Global refinery runs outside the Middle East declined by 1.4 mb/d m/m in March, from 75.5 mb/d to 74.1 mb/d, and are expected to decline by another 3.1 mb/d m/m in April to close to 71 mb/d, pushing runs 4.5 mb/d below pre-war February levels. Some countries' decision to restrict exports of refined products via bans (e.g., China) or higher taxes (e.g., India) reduced further the availability of refined products. The impacts of this 'perfect storm' have been felt most strongly on diesel and jet fuel with refining margins and diesel and jet fuel prices reaching record levels, sending signals for consumers to ration demand and for refineries to boost runs (**Figure 6**).

[9] Financial Times. 'Asian refineries slash output as Iran crisis chokes crude supplies', 24 April 2026.

2.3. From quantifying losses to assessing the recovery phase

At the early stages of the war, the market's focus was on quantifying output losses. But as the disruption persisted, the market broadened its focus towards estimating the time needed to restore production to pre-war levels.

A return to pre-war output levels would take months and is unlikely to be uniform across countries. Restart timelines are not straightforward and will vary according to whether shut-ins were full or partial, whether affected assets are predominantly onshore or offshore, and whether any physical damage has occurred. They will also depend on technical changes across the wider production system during the shut-in, including in reservoirs, wellbores, flowlines, processing facilities, and export routes. Critically, longer shut-ins increase restart difficulty through four main channels at once: reservoir changes, deposition and fluid redistribution, flow assurance problems, and a more complex facilities/export recommissioning process, thereby increasing the risk of a slower, less complete, and more operationally challenging recovery.[10]

In some cases, longer shut-ins can also lead to permanent capacity loss where the interruption damages wells, reservoirs, or critical surface infrastructure.[11] The risk is amplified in producers with large integrated offshore and onshore systems, where a prolonged outage can impair the wider production chain, so that the resulting loss of capacity is no longer purely temporary but may become structural, especially in mature or technically complex fields.[12] This risk is especially relevant in the Gulf because some producers combine very large onshore systems with complex offshore and subsea networks (e.g. the UAE and Qatar), so that a prolonged shut-in can affect not only individual wells but also the wider system of flowlines, processing facilities and export routes, while in others, it is compounded by mature-field and infrastructure constraints (e.g. Iraq).

In terms of individual countries (see **Table 2**), **Saudi Arabia** is best placed to achieve the fastest and most complete recovery to pre-war output levels, as it combines very large technical and financial resources with demonstrated restoration experience and the most meaningful bypass flexibility through the East-West pipeline. In fact, following the mid-September 2019 attacks on the Abqaiq and Khurais facilities, which damaged infrastructure and led to the shut-in of 5.7 mb/d of Saudi production, Aramco restored supply of around 10 mb/d within 10 days and aimed to regain its maximum capacity of 12 mb/d within less than two months.[13] The **UAE** is also well placed, combining strong operational capability with bypass export capacity via Fujairah, although it is somewhat more exposed than Saudi Arabia if disruptions heavily affect ADNOC's offshore fields, which account for around half of total oil production capacity. This is also likely to delay ADNOC's offshore ambitions to push offshore Upper Zakum capacity above 1.5 mb/d and total UAE capacity above 5 mb/d by 2027. A similar consideration applies to **Qatar**, whose broader production system, despite strong technical capability, relies heavily on offshore output linked to onshore processing. **Kuwait** is in a reasonably good position because the bulk of its production is concentrated onshore, which should make restart and recovery less complex, but it remains highly dependent on the normalization of flows through the SOH given its lack of bypass options. **Iraq** is in the weakest recovery position for the same reason, compounded by infrastructure and export bottlenecks. The shut-ins are also impacting refinery operations in Iraq. Iraq's Oil Products Distribution Company has noted recently a drop in octane quality from North Refineries Company[14] and a decrease in the flash point for diesel, further compounding stresses on the country's economy.

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[10] JPT. 2020. 'Reservoir Commentary: Potential Implications of Long-Term Shut-Ins on Reservoir', Commentary by the *SPE Reservoir Advisory Committee* (RAC), 27 May.

[11] This is most likely when prolonged shut-ins allow plugging, deposition, corrosion, or broader near-wellbore impairment to reduce productivity, so that production can only by restored through workovers or equipment replacement and may still not return to pre-shut-in levels.

[12] Garduno, J. and King, G. 2020. Managing Risk and Reducing Damage From Well Shut-Ins, Journal of Petroleum Engineering (JPT), 29 April.

[13] Reuters. 'Aramco hopes to repair remaining damage from attacks by end-November', 12 October 2019.

[14] Refineries Company owns Iraq's largest refinery, Baiji.

Table 2: Middle East Gulf oil production recovery outlook, in mb/d

	Iraq		Kuwait		Qatar		Saudi Arabia		UAE		Non-crude*		TOTAL
Jan-26	**4.16**		**2.58**		**1.27**		**10.09**		**3.39**		**4.67**		
Feb-26	**4.19**		**2.58**		**1.25**		**10.88**		**3.42**		**4.69**		
Mar-26	1.62	(2.56)	1.21	(1.37)	0.18	(1.07)	6.96	(3.92)	1.89	(1.53)	3.03	(1.66)	**(12.11)**
Apr-26F	1.39	(2.80)	0.77	(1.82)	0.10	(1.15)	7.46	(3.42)	1.67	(1.75)	2.96	(1.73)	**(12.66)**
May-26F	1.81	(2.37)	1.31	(1.28)	0.33	(0.92)	9.58	(1.30)	2.15	(1.27)	3.68	(1.01)	**(8.15)**
Jun-26F	2.53	(1.66)	1.70	(0.88)	0.57	(0.68)	10.23	0.00	2.92	(0.50)	3.97	(0.72)	**(4.45)**
Jul-26F	3.25	(0.94)	2.09	(0.49)	0.74	(0.51)	10.23	0.00	3.20	(0.22)	4.15	(0.54)	**(2.69)**
Aug-26F	3.52	(0.67)	2.35	(0.23)	0.84	(0.41)	10.23	0.00	3.45	0.00	4.21	(0.48)	**(1.79)**
Sep-26F	3.69	(0.50)	2.48	(0.10)	0.94	(0.31)	10.23	0.00	3.45	0.00	4.33	(0.36)	**(1.26)**
Oct-26F	3.69	(0.50)	2.61	0.00	0.94	(0.31)	10.23	0.00	3.45	0.00	4.35	(0.34)	**(1.15)**
Nov-26F	3.69	(0.50)	2.61	0.00	0.94	(0.31)	10.23	0.00	3.45	0.00	4.35	(0.34)	**(1.15)**
Dec-26F	3.69	(0.50)	2.61	0.00	0.94	(0.31)	10.23	0.00	3.45	0.00	4.35	(0.34)	**(1.15)**

Notes: *Non-crude includes NGLs and condensates excluding Qatar condensates. Assumes a gradual normalization of Hormuz flows from May onwards.
Source: OIES

As in the case of oil production, the duration of refining outages is also uncertain and would vary across countries depending on many factors including the damage incurred by the various facilities, the state of the infrastructure, and how quickly trade will normalize through the SOH.

2.4. Who has sovereignty over the Strait of Hormuz?

For all the Gulf countries, the recovery path to restoring production to pre-war levels will depend primarily on the normalization of flows through the SOH. The immediate aftermath of the US-Israeli strikes on Iran on 28 February 2026 saw Iran move to restrict maritime traffic and effectively 'close' the SOH. Following unsuccessful negotiations between the US and Iran, the US also imposed its own blockade.

Iranian pressure over Hormuz has a long genealogy. It developed first as a legal-sovereignty challenge during the UNCLOS III (Third United Nations Conference of the Law of the Sea) negotiations of the 1970s, when Iran and Oman resisted an unrestricted transit-passage regime in the Strait.[15] Second, as operational interference during the 1980s 'Tanker war', when Iran disrupted Gulf shipping without imposing a sustained total closure.[16] Even so, the literature does not document a prolonged total closure of Hormuz. Rather, it assumes any outright closure to be both unlikely and difficult to sustain for long (estimates ranging from a few days to several weeks) in light of Iran's own dependence on the SOH and the likely US-led military response to reopen it.[17]

Iran has realised early on during this conflict that restricting flows through SOH is one of its strongest negotiating cards and one that could potentially deter future attacks. Thus, it has taken measures to formalise its control over the SOH through imposing a toll for 'its services' and putting in place a system

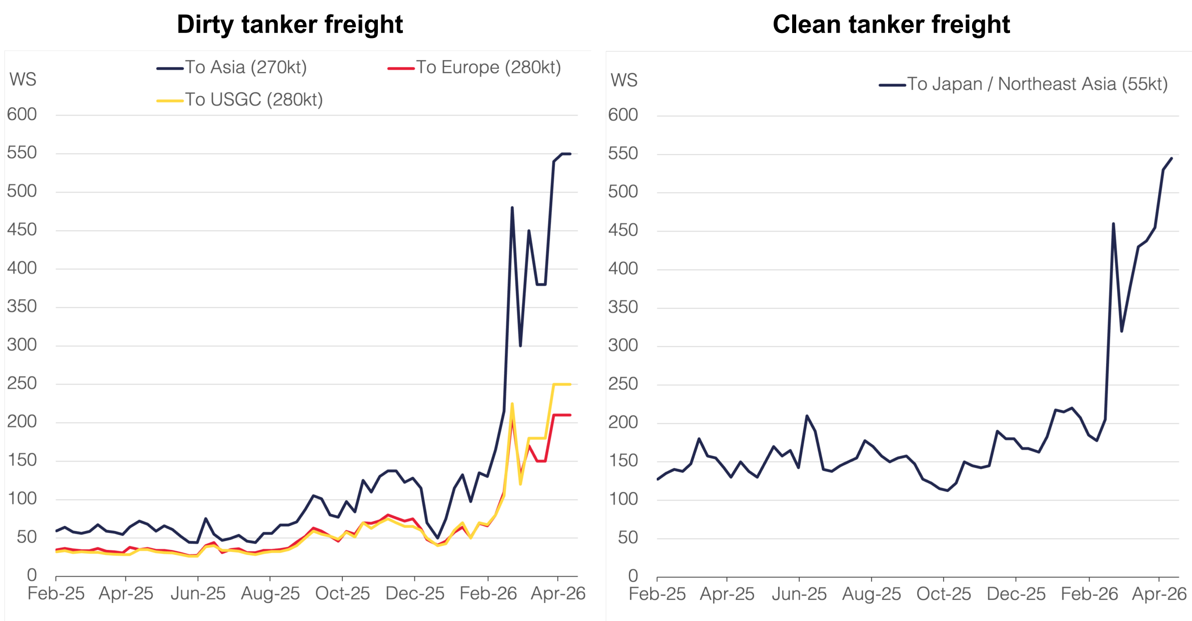
[15] Franckx, E. and Razani, A. 1974. The Strait of Hormuz, In *The Proceedings of the Symposium on the Straits Used for International Navigation*, Vol. 28, p.53.
[16] Katzman et al. 2012. Iran's Treat to the Strait of Hormuz, No. R42335.
[17] Ratner, M. 2018. Iran's threats, the Strait of Hormuz, and oil markets: In brief, *Congressional Research Service*, CRS report prepared for members and committees of congress, Aug 6.

where verification and permission are required for passage. The US position was less clear and kept shifting from calling for safe and free navigation, to implementing a joint toll system with Iran, to considering the opening of SOH as a problem that other countries should address.

A fundamental shift occurred after the failed negotiations in Pakistan in early April with the US announcing the implementation of its own blockade focused on vessels departing or heading towards Iranian ports. This reduced Iran's leverage while increasing its cost as Iran itself remains heavily reliant on exports and imports through SOH. In response, Iran has threatened that it would retaliate by attacking ports in the Gulf. China, which so far has managed to arrange for a few of its vessels to pass through the SOH, called the US blockade 'irresponsible and dangerous'.[18] Europe has plans to form a broad coalition to secure shipping and 'restore freedom of navigation via a 'multilateral and purely defensive mission' and 'when the security conditions allow it'.

Fundamental questions facing the oil market are not only when the flows through SOH will normalise, but also whether 'safe and toll-free freedom of navigation in the Strait of Hormuz, consistent with UNSC Resolution 2817 and the Law of the Sea' will ever be restored. The uncertainty surrounding the future framework governing the passage through the SOH represents one of the key unintended consequences of the current war.

Fig. 7: Mideast Gulf tanker freight rates



Notes: Worldscale (WS). Data shown are weekly, as of week-ending 24 April 2026. Source: Argus

Regardless of the outcome, the risk of transit through the SOH has materially risen increasing the cost of oil trade via higher insurance premiums and freight rates, which could take months to normalise to pre-war levels (**Figure 7**). Higher costs could also include toll payments, more complex logistics, new verification and inspection systems and increased security costs, for instance, if passage is guaranteed by the US or a coalition force.

And if the 'safe and toll free freedom of navigation' is not restored, then this will have some important implications for oil markets as it may result in the emergence of a multi-tier system whereby some vessels secure free passage through bilateral negotiations, others are permitted to pass but are subject to the proposed toll system, potentially in currencies other than the US dollar and some vessels are

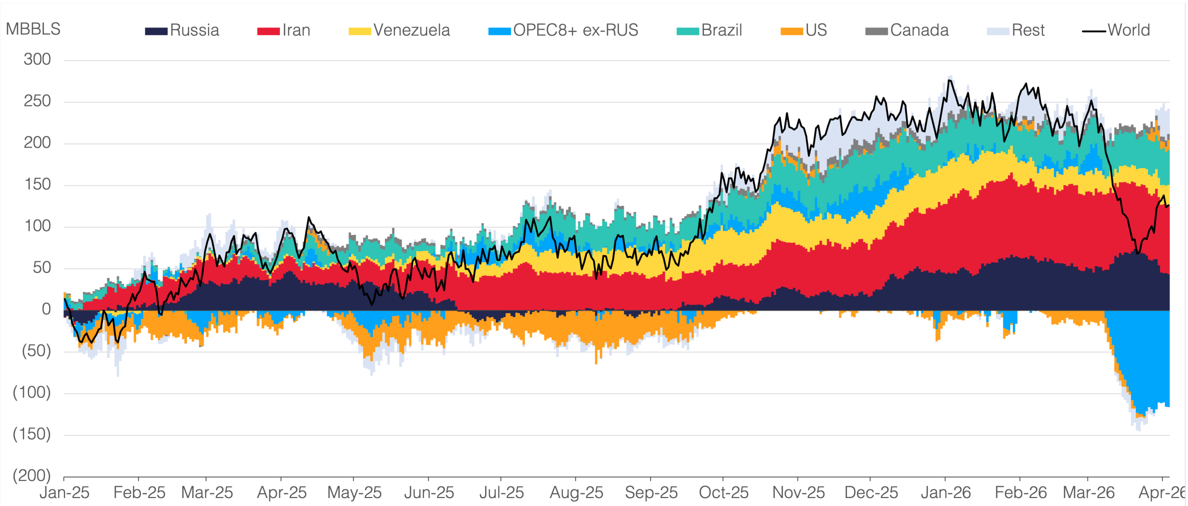
[18] Reuters. 'US, Iran may resume talks this week despite port blockade', 15 April 2026.

denied passage altogether. In such a multiple tier system, trade flows are likely to become more volatile resulting in higher oil market volatility. Also, key Gulf oil exporters may decide not to transit under a framework where access is 'restricted, conditioned and controlled' and invest in new pipelines that bypass the SOH altogether (discussed in more detail below). Also, some buyers may request that Gulf crude is sold on delivered basis to offset the higher cost and risk premiums with large implications on the structure of Middle East sales and pricing.

3. Buffers in the oil system are thin

The volumetric buffers in the oil system available to mitigate such a severe supply shock have been eroded in large part due to the unavailability of spare capacity, which is mostly concentrated in the GCC, the same region suffering from the disruption. Unlike commercial and strategic stocks, spare capacity represents a 'flow' buffer as countries with spare capacity can increase their production and maintain it at that level. In contrast, commercial and strategic stocks are limited. To put things in perspective, IEA member countries agreed to release 398 million barrels from strategic petroleum reserves (SPR). Yet, by the end of April, only two months after the start of the war, cumulative output losses are projected to reach around 790 million barrels – almost twice the planned release. Thus, while the strategic reserves can act as a relief, it is short lived: the stock release is quite small compared to the size of the shock and can't be maintained beyond few months depending on the rate at which these stocks are being released. In fact, the planned total IEA release of 398 million barrels includes 273.5 million barrels of crude, while the remaining 125 million barrels consist of refined products. This crude-products distinction matters because only the crude portion directly addresses the crude supply gap through refinery feedstock availability and broader market rebalancing. Refined product releases, by contrast, mainly alleviate tightness in end-use fuel markets and help mitigate the demand response under supply tightness. Also, there is a quality mismatch and regional mismatch as most of the production loss is medium sour and stocks are not released in places where they are most needed and arbitraging these barrels takes time and is costly, especially given the high freight costs. Part of the US SPR release of medium crudes has been sold to refineries in Europe and could potentially find its way to Asia[19].

Fig. 8: Observed global oil-on-sea change in 2025 and 2026, cumulative, Dec 24 = 0

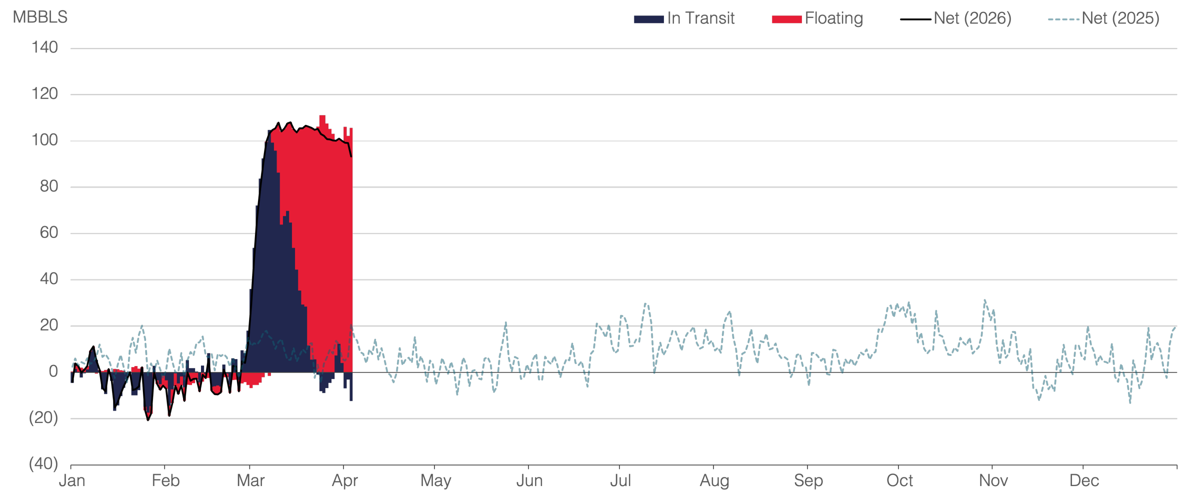


Notes: Data shown from 1 January 2025 to 3 April 2026. Includes crude and oil products.
Source: Kpler, OIES

[19] Bloomberg. 'Trump's Emergency Oil Sails to Europe as War Upends Energy Flows', 23 April 2026.

Other buffers in the system included sanctioned Iranian and Russian barrels in floating storage. By mid-February, Kpler estimates show that these volumes had risen to 50.85 million barrels, their highest level since records began in 2016. At the start of the war, the US granted waivers for Iranian and Russian sanctioned crude in floating storage, and while initially the US Treasury announced that it will not extend the waivers, it shifted its position and extended the waiver for Russian oil loaded on or before April 17.[20] Volumes of crude on the water, which includes oil in transit and in floating storage, were central to the oil glut narrative and the view that the oil market is well prepared to withstand a supply shock. However, this relief proved to be short-lived and crude on the water (both oil in transit and in floating storage) has now fallen sharply from 2045.9 million barrels end-February to 1944.9 million barrels end-March. This unwound around half of the large build-up accumulated in 2025 (-51%), with surplus oil-on-sea falling to 126.2 million barrels by the end of March, from 255.4 million barrels at end-2025 (see **Figure 8**). This occurred despite a large increase in floating storage, due mainly to stranded barrels in the Middle East Gulf following the closure of the SOH, which were estimated at 110.4 million barrels by end-March, up from 9 million barrels at end-February (**Figure 9**).

Fig. 9: Observed oil-on-sea change in Mideast Gulf in 2026, cumulative, Dec 25 = 0



Notes: Data shown from 1 January 2025 to 3 April 2026. Includes crude and oil products.
Source: Kpler, OIES

In short, given the magnitude of the production loss and the unavailability of spare capacity, the buffers in the system are insufficient to replace the volumes of lost barrels, and if the disruption persists for longer, the existing buffers will become less effective. Consequently, prices will have to play a central role in the adjustment through their impact on production and consumption. Given that oil production is highly inelastic in the short term, demand is likely to carry the bulk of adjustment through a combination of price and income effects and policy measures as well as demand rationing. In April, both the IEA and the US EIA revised lower their oil demand growth forecasts for 2026, by 730 kb/d and 600 kb/d respectively.[21,22] The IEA now projects a small demand contraction of 80 kb/d this year – the first annual decline since the COVID shock in 2020 – while the US EIA still expects demand growth of 600 kb/d.

Two issues complicate the prospect of a large demand response. First, the empirical literature consistently finds short-run oil demand to be relatively price inelastic, with estimates often clustered around -0.1.[23] This means that, given the magnitude of the supply shock, price would need to move

[20] Bloomberg, 'US Again Allows More Russian Oil Sales to Help Control Prices', 18 April 2026.
[21] IEA. Oil Market Report, 14 April 2026.
[22] US EIA. Short-Term Energy Outlook, 7 April 2026.
[23] Caldara, D., Cavallo, M. and Iacoviello, M. 2016. 'Oil Price Elasticities and Oil Price Fluctuations', *Journal of Monetary Economics*, 103, pp. 1-20.

very sharply to generate even a partial offsetting demand response in the short-run. For example, using global oil demand of around 100 mb/d, the 11.5 mb/d m/m global supply loss estimated in March would require demand to fall by roughly 11.5% to offset the shock fully. At an assumed short-run price elasticity of -0.1, this implies a price increase of around 115% under a simple linear elasticity approximation. By comparison, Brent prices rose by 46% in March, from $71.1/b to $103.7/b, and they are likely to average slightly lower in April despite the supply shock persisting and even deepening. Moreover, even this estimate is likely conservative, as short-run demand elasticities usually capture adjustment over months or quarters rather than within a single month. This suggests that the observed price response has so far been insufficient to generate the scale of demand adjustment required, while delayed pass-through and adjustment lags limit the speed at which price can help clear the market.

Second, the sheer size of the current supply shock would require a demand response of comparable magnitude, one rarely seen in modern oil market history. The closest recent analogue is the COVID shock in 2020, when a 10.1 mb/d m/m collapse in global oil demand was accompanied by a large 12.1 mb/d downward adjustment in global supply over the following months, most notably through OPEC+ output cuts, which accounted for more than 80% of the total supply adjustment. The adjustment mechanisms, however, are fundamentally different. In 2020, the demand shock was largely policy-driven, reflecting mobility restrictions and the broader shutdown of economic activity, while a large part of the supply response was voluntary through OPEC+. In the current crisis, supply buffers are limited, while the 2020 demand-side adjustment mechanisms are not realistically available and cannot be replicated without imposing severe economic and social costs. Given also the limited size of the direct price response, this makes the macroeconomic/income channel an increasingly important mechanism for generating a larger demand response, although its transmission is typically slower, working through weaker GDP growth, lower real incomes, tighter financial conditions, and reduced industrial and transport activity. So far, Oxford Economics has revised its 2026 global GDP growth forecast lower by around 0.6 percentage points, from 3.0% in February to 2.4% in April.

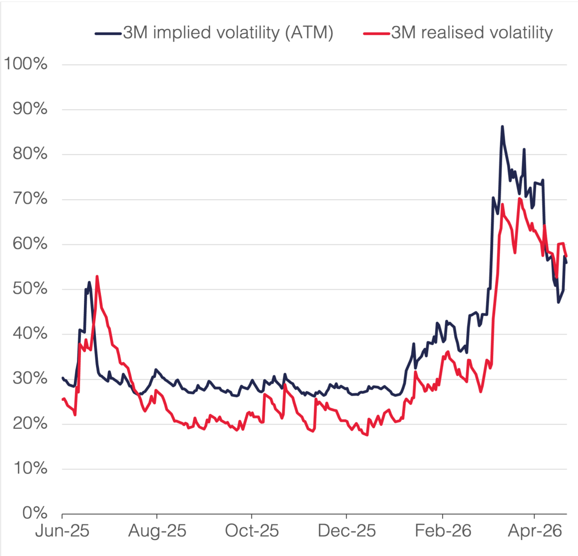
4. The physical shock propagates into the financial layers

The current supply shock is occurring in the context of a highly financialized and futurized oil markets. The financial layers play a key role in price discovery, connecting different oil markets and benchmarks across geographies and time, and enable market participants to manage their risks and take speculative positions. The flows through these financial layers are larger than the underlying physical oil consumption and production and are driven by various factors including changes in oil market fundamentals, expectations about these fundamentals, technical factors, and the regular flow of information. They are also driven by spillovers to the oil market from other much larger financial markets such as equity and bond markets.

The current supply shock has propagated through the financial layers impacting oil price movements, volatility and liquidity. Implied and realised volatility in key benchmarks such as Brent and WTI and key products such as ICE gasoil have spiked to record levels (**Figure 10**). Higher volatility alongside higher margin requirements by futures exchanges made near months almost untradable especially for small players. Higher uncertainty and volatility attracted new set of financial players into the options markets. Option volatility skews titled heavily towards calls as supply losses increased the demand for insurance against higher prices, while in normal times these are usually tilted towards puts due to hedging demand from producers.[24] While the demand for call options surged, the ability of option dealers to meet that demand was limited amplifying the volatility skew (see **Figure 11**) and forcing them to buy futures. This encouraged further momentum trading.[25]

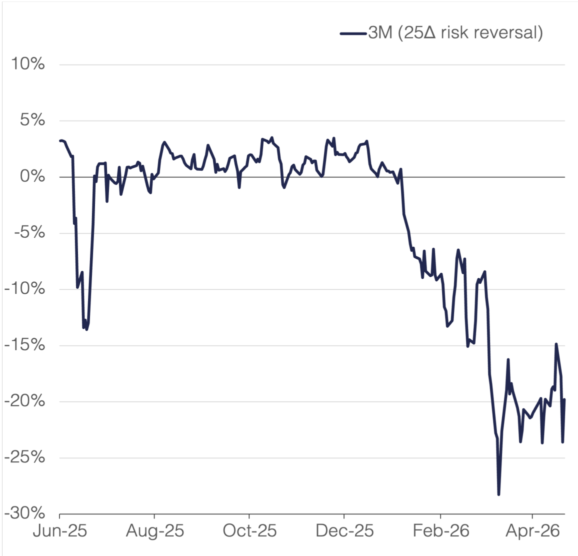
[24] Bouchouev, I. 2026. 'Energy Quantamentals: The Oil Crisis in the Eyes of a Financial Trader', OIES Comment, April.
[25] Bouchouev, I. 2026. 'Energy Quantamentals: The Oil Crisis in the Eyes of a Financial Trader', OIES Comment, April.

Fig. 10: ATM implied and 21-day realised volatility (Brent)



Source: Bloomberg, OIES

Fig 11: 25-delta risk reversal (puts vs calls)

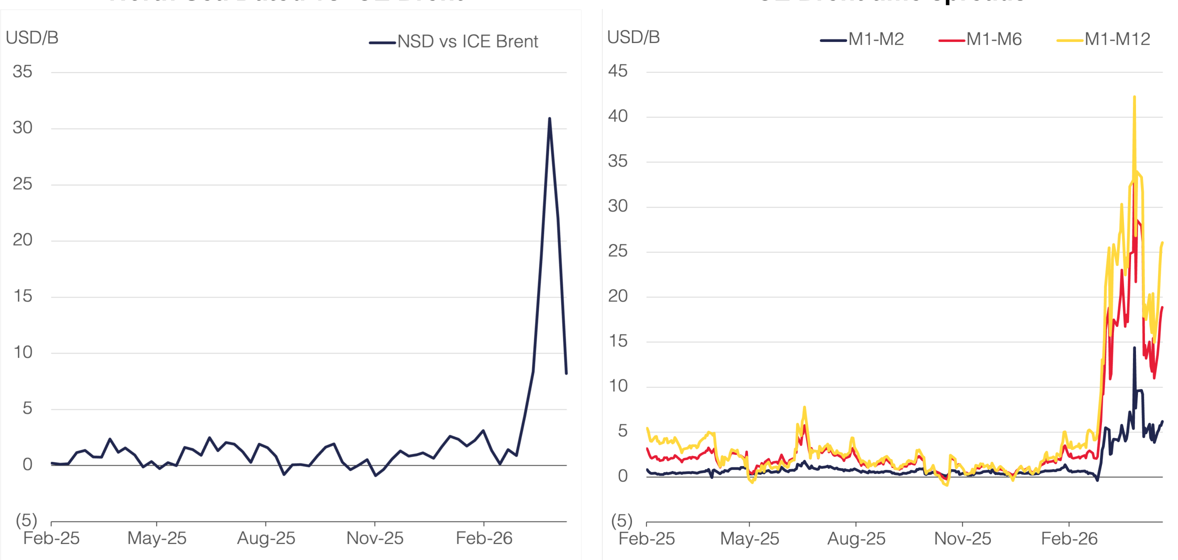


Source: Bloomberg, OIES

The heightened volatility was also reflected in the various spreads, reflecting the different pressure points in the oil system (physical versus financial), over which time horizon (10-30-day delivery versus longer delivery) and location (FOB versus delivered). The Dated Brent premium to front-month ICE Brent traded at a record level of around $31/b in the week ending 10 April, reflecting the scarcity of physical barrels, especially those available for prompt delivery (**Figure 12**). ICE Brent time spreads also moved to extreme levels, widening sharply across the curve (see **Figure 12**). Refining margins for certain products such as diesel and jet fuel reached record levels exacerbated by refineries closing their paper positions as their physical positions were compromised.[26]

Fig. 12: Brent crude spreads

North Sea Dated vs ICE Brent | **ICE Brent time spreads**



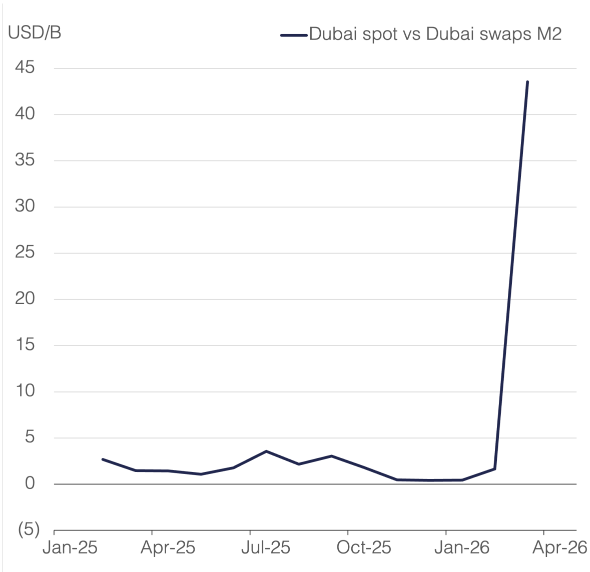
Notes: Data shown as of week-ending 24 April 2026. Source: Argus, Bloomberg, OIES

[26] Bouchouev, I. 2026. 'Energy Quantamentals: The Oil Crisis in the Eyes of a Financial Trader', OIES Comment, April.

These sharp movements alongside losing access to physical barrels have resulted in some players incurring huge trading losses in OTC markets. These losses were particularly acute in the East of Suez where players usually take short positions in paper markets which are offset by taking long positions on physical. With many participants losing access to physical barrels, this exposed them to huge losses as they had to prematurely close their financial position at a much higher price.[27]

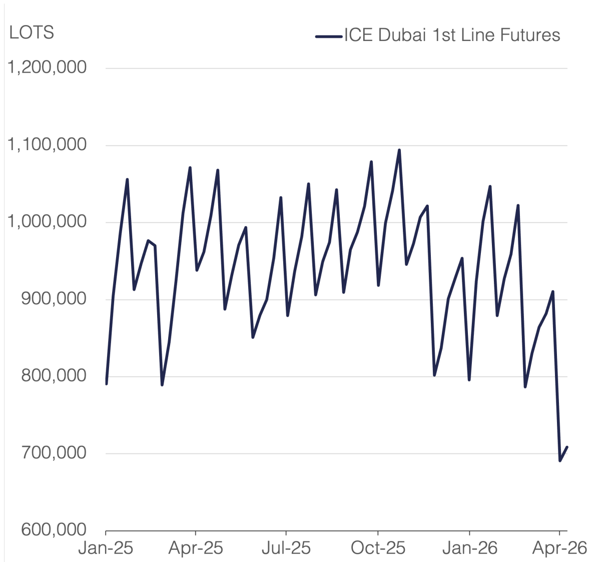
The supply shock also occurred against a background of a more varied pricing system and bigger role for futures exchanges in price formation, particularly in the Gulf with the launch of futures contracts such as GME Oman and IFAD Murban and the availability of destination-free barrels that could be treaded in Platt's Market on Close (MOC)[28]. These benchmarks saw some erratic movements as liquidity dried up (**Figure 13**). The impacts were highly visible in some markets such as Dubai, with open interest in ICE Dubai open declining sharply (**Figure 14**). These price moves were exacerbated by some players' decisions such as S&P Global Platts announcement that it will no longer accept nominations by sellers for some of the grades such as ALS and Upper Zakum and ADNOC's decision to cut supplies to equity lifters, which are key liquidity providers.[29] Lower liquidity amplified volatility and increased the pricing power of some players impairing the price discovery process and exacerbating some of the players' losses. This caused some Asian refineries to request that producers change their crude pricing system.[30] The current trade cycle in Asia has already seen a shift away from Dubai as the basis for pricing OTC deals in favour of Dated Brent.[31]

Fig. 13: Dubai physical differentials



Notes: Data shown are monthly, from February 2025 to March 2026. Source: Argus, OIES

Fig 14: ICE Dubai open interest



Source: ICE

5. Key player's responses to the shock

The impacts of the oil supply shock have been profound, forcing some fundamental shifts in the position of key oil players and shaping their responses. Below is a short description of the position of key players.

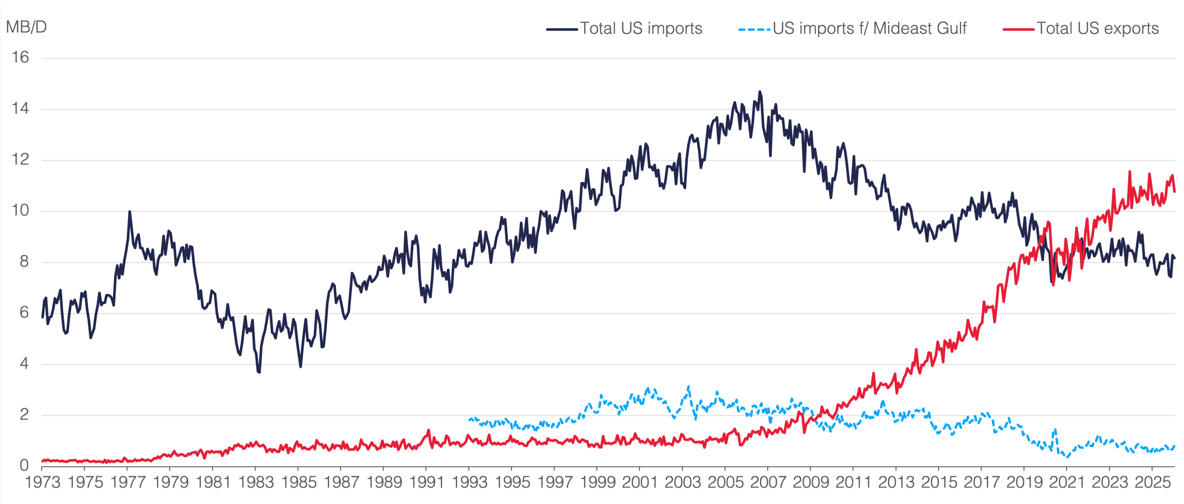
[27] Fattouh, B. and Mehdi, A. 2026.'Through the Looking Glass: Oil and the Search for Direction', OIES Comment, March.
[28] Fattouh, B. and Mehdi, A. 2025.'Dubai, we have a problem: Murban and Middle East crude pricing', OIES Comment, May.
[29] Fattouh, B. and Mehdi, A. 2026.'Through the Looking Glass: Oil and the Search for Direction', OIES Comment, March.
[30] Bloomberg. 'Asia Refiners Ask Saudis to Change Oil Price System Amid War', 19 March 2026.
[31] Renaissance Energy. Oil Monthly, April 2026 Issue.

5.1. US: The limits of the US Energy Dominance Doctrine

The concept of energy dominance has become a defining feature of President Trump's energy policy. Though it is difficult to define, energy dominance seems to emphasise three key aspects: energy abundance in the form of higher production, low-cost energy and higher exports. These aspects provide the US with more flexibility in its foreign policy including to 'compete with hostile foreign powers'.[32]

Since 2005, US crude imports have been in decline in large part due to the sharp increase in US production (see **Figure 15**), which rose by 8.4 mb/d between 2005 and 2025, from 5.2 mb/d to 13.6 mb/d. Rising US production alongside higher imports from Canada reduced the share of crude oil imports from the Gulf in overall imports, which stood at less than 1% in 2025. Thus, in term of physical supplies, the US does not face the risk of physical shortages.

Fig. 15: US imports and exports of crude oil and petroleum products



Notes: Data shown from January 1973 to January 2026.
Source: US EIA

In fact, one of the effects of the war has been the increase in global demand for US crude. The Russian-Ukraine war already boosted WTI's role in global trade flows. The current war further enhanced WTI's role in oil markets and price formation. As buyers scramble for barrels, US crude exports have risen sharply in March and April and are poised to reach a record level of 5 mb/d in May.[33] WTI in Europe has been trading at a large premium as competition for US crude has intensified particularly from Asia. This reinforced WTI Midland's role as the world's most important swing grade and shows Asia's growing interconnectedness to US pricing. However, there are limits as to how much US crude exports could rise due to production and export infrastructure constraints.

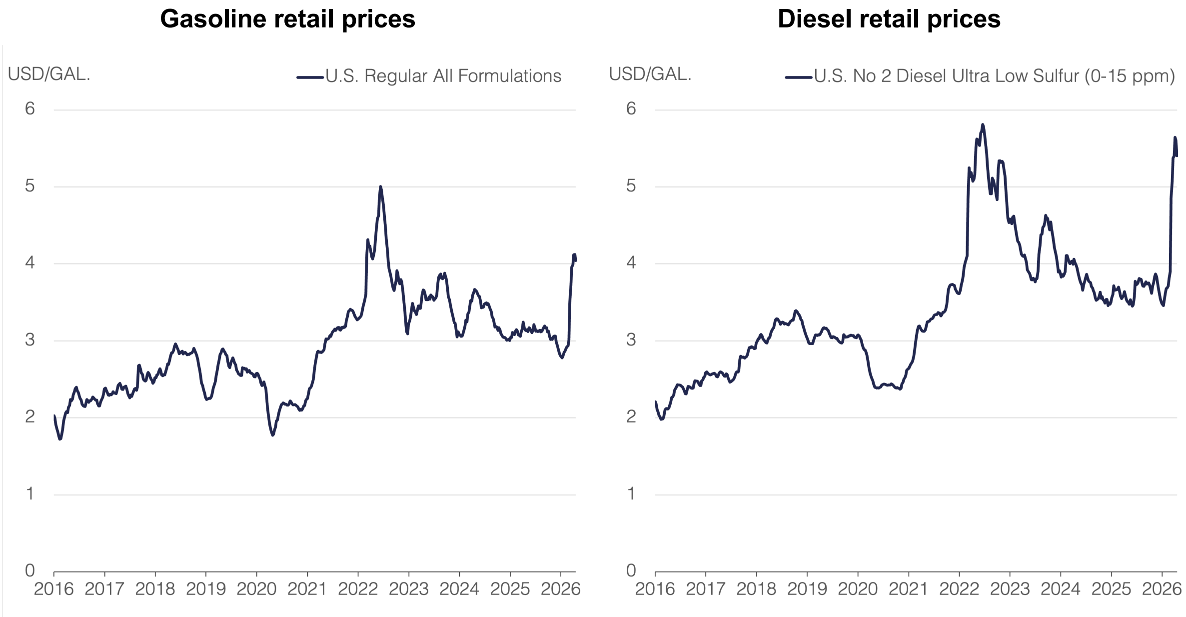
The increase in exports was not only limited to crude. US exports of products especially diesel and jet fuel have also risen to record levels, with US refineries capturing larger margins given their access to lower priced domestic crudes.[34] The disruption of LPG through the SOH (around 30% of global seaborne exports passed through the SOH in 2025) also increased demand for US LPG with exports rising by nearly 15% m/m in April 2026 to an all-time high of 2.7 mb/d. But as in the case of crude, there are limits to products' exports: surplus production of diesel and gasoline in the US already has its

[32] The White House (WH). "Presidential Determination Pursuant to Section 303 of the Defence Production Act of 1950, as Amended, on Natural Gas Transmission, Processing, Storage, and Liquified Natural Gas Capacity', Memorandum for the Secretary of Energy, 20 April 2026.
[33] Bloomberg. 'US Oil Exports to Hit 5 Million Barrels a Day Amid Global Crunch', 9 April 2026.
[34] Financial Times. 'US oil refiners reap windfall from Iran war', 20 April 2026.

customers, and with no additional refining capacity and with refining runs falling in recent weeks, the US is unlikely to make more products available for Asia.

When it comes to the cost aspect, the current supply shock has shown that the US can't fully insulate itself from the effects of higher global oil price as it filters through to consumers and the broader economy through higher gasoline prices, inflation and inflationary expectations.[35] Despite the Trump Administration's attempts to control the upward movement in the oil price including intervening in the oil market heavily in terms of information flows and messaging, granting a temporary waiver of the Jones Act, releasing oil from the SPR under an exchange programme, and granting sanctions waivers on Russian and Iranian oil in floating storage, gasoline prices have increased from less than $3/gallon to above $4/gallon and US inflation has risen by 3.3% in March (from a forecast of 2.5% before the war), driven largely by the sharp jump in petrol prices (gasoline and diesel; see **Figure 16**).[36]

Fig. 16: US retail gasoline and diesel prices



Notes: Data shown are weekly through 20 April 2026.
Source: US EIA

Measures to incentivise US producers to increase production have been largely ineffective. Although US shale investment cycle is relatively short, the lags between investment and production increases take months and given the high level of uncertainty, US shale companies may adopt a wait and see approach and defer increasing capex until clearer signals emerge. So far there is not much evidence of sharp increase in activity in the US shale patch. The US rig count has barely moved, with active rigs in the Lower 48 states standing at 529 in March 2026, only one rig higher than in December 2025, prompting the Trump administration to urge oil and gas companies to increase drilling and output.[37] According to preliminary US EIA estimates, by March 2026 US crude production declined from its peak of 13.86 mb/d in September 2025 to 13.56 mb/d in March 2026, a fall of 300 kb/d. That being said, projections of US production for 2026 and 2027 have been revised upwards. At the end of 2025, OIES

[35] Financial Times. 'Impact of Iran war will hurt US even after conflict ends, economists warn', 19 April 2026.
[36] Financial Times. 'Impact of Iran war will hurt US even after conflict ends, economists warn', 19 April 2026.
[37] Bloomberg. 'Trump Officials Urge Oil Industry to Boost Output Amid War, 17 April 2026.

projected US crude production growth of only 40 kb/d for 2026;[38] this has since been revised modestly higher to 180 kb/d, with growth of 100 kb/d expected in 2027.[39]

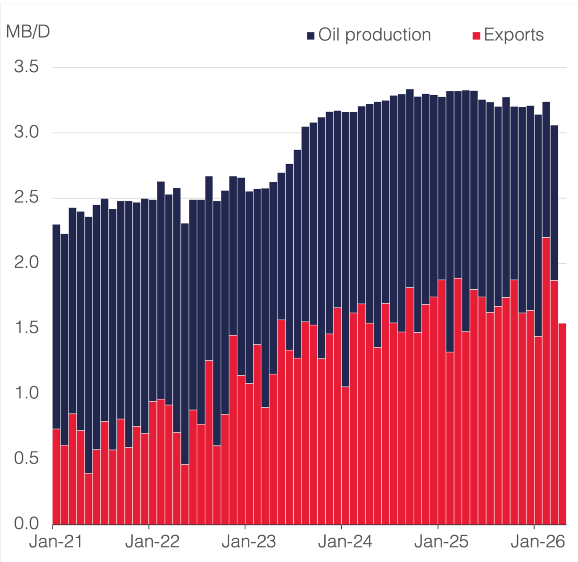
In short, while higher oil and gas production and exports have placed the US in a better position to mitigate the impacts of the oil shock and has generated a large windfall for US producers, the current war still exposed the limits of US energy dominance. The interconnectedness of oil markets means that the US can't insulate itself from the impacts of higher and more volatile oil prices, and this in a way acted as a constraint on US strategy towards Iran. Although the US exports of crude and products have risen, there are limits to further increases due to infrastructure, production and refining constraints.

5.2. Iran: Cementing its leverage over the Strait of Hormuz

As part of its broader strategy to impose maximum pain on the global economy, Iran has attacked energy infrastructure in the Gulf and disrupted free passage through the SOH. While Iran has not 'formally' blocked the Strait, it has shown its readiness to weaponize the SOH by attacking tankers, requesting fees in currencies other than the US dollar, and imposing conditions on free passage, causing safety and security issues, and prompting insurance premiums to rise sharply and for war risk cover to be withdrawn altogether.

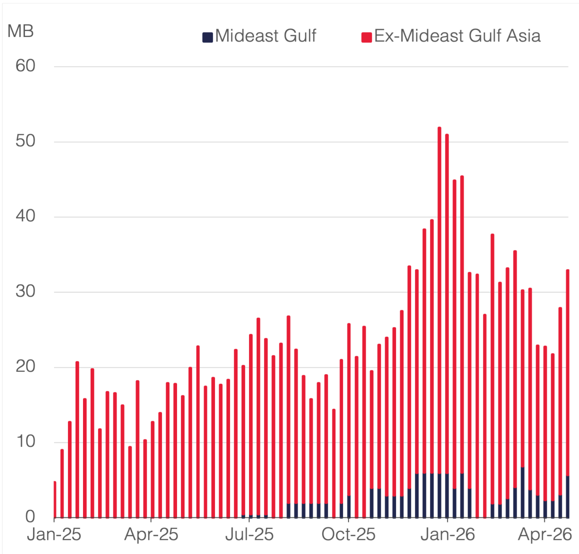
Iran has been quick in formalizing the control over the SOH through a draft law that establishes Iran as the key guardian of the SOH including introducing an inspection and verification system, a tolling system where vessels pay a transit fee in riyals or cryptocurrency, with the power to block 'unfriendly' vessels from passing through the SOH. Several Asian countries have concluded bilateral agreements for ships to pass safely through the SOH. As discussed above, in effect, this will create a multiple tier system where some countries negotiate safe passage of its vessels with or without fees, while others are prevented from passing through the SOH, and/or refuse to transit under a framework where access is 'restricted, conditioned and controlled'. It is quite telling that when Iran recently declared the SOH is open as part of the ceasefire deal, this was conditional on using 'the coordinated route as already announced by Ports and Maritime of the Islamic Republic of Iran'. This does not meet the definition of 'free passage' and clearly indicates that Iran's control over the SOH will be a major item in the US-Iran negotiations.

Fig. 17: Iran supply



Source: OPEC Secondary sources, Kpler, OIES

Fig 18: Iran floating storage in Asia



Source: Kpler, OIES

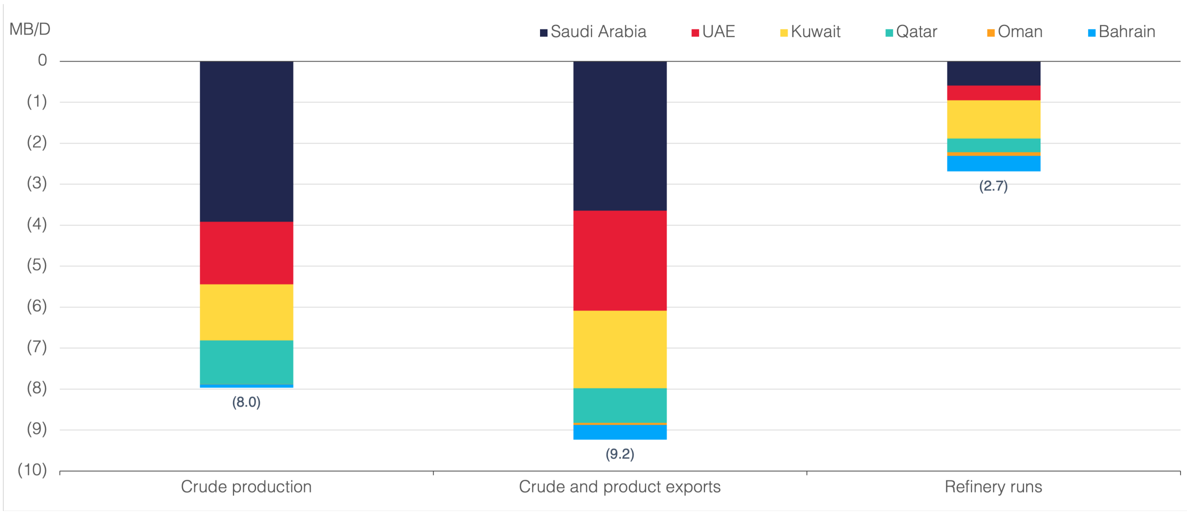
[38] OIES. OIES Oil Monthly Issue 50, 19 December 2025.
[39] OIES. OIES Oil Monthly Issue 53, 23 April 2026.

With the US imposing its own blockade, this has reduced Iran's leverage and Iranian exports, which have been largely maintained in March, are now projected to decline sharply (see **Figure 17**). If implemented successfully, choking off exports would force Iran to cut its oil production though there are different views on the timing frame and whether the shut-ins would cause permanent damage to Iranian fields. Iran can still temporarily boost its oil revenues by selling oil from floating storage in vessels anchored in ex-Middle East Gulf Asia which Kpler estimates at around 27 million barrels (**Figure 18**).

5.3. The GCC: Preserving the role of a reliable supplier and spare capacity holder

Since the start of the war, GCC crude production has fallen from 19.1 mb/d in February to 11.1 mb/d in March (-42%), exports of crude and products from 19.2 mb/d to 10 mb/d (-48%) and refining runs from 6.2 mb/d to 3.5 mb/d (-43%; see **Figure 19**). Also, key GCC energy infrastructure has been damaged with Rystad Energy estimating repair and restoration costs at $34-58 billion, the majority of which relates to oil and gas facilities, estimated at up to $50 billion.[40] This has taken a toll on some of the GCC economies. In April, the IMF revised down the GCC growth outlook by 2.3 percentage points.

Fig. 19: GCC supply losses by origin, Mar-26 vs Feb-26



Source: OPEC Secondary sources, IEA, Kpler, OIES

Beyond these direct impacts on their oil sector and the economy, any framework that restricts free passage through the SOH undermine GCC's role as a reliable and secure oil and gas supplier and the holder of spare capacity. This was recently highlighted by the CEO of ADNOC who stated that 'the Strait must be open, fully, unconditionally and without restriction. Energy security and global economic stability depend on it'.[41]

If one of the outcomes of the war is that free passage can no longer be guaranteed, this could prompt key GCC producers to reroute exports away from the SOH. Existing bypass routes in Saudi Arabia and the UAE already demonstrate the strategic value of such alternatives. However, investments in pipelines are costly, take a long time, need to meet the commerciality criteria, require changes in marketing, trade and pricing patterns, and would not necessarily offer full security as these can also be subject to attacks. Given the long lead times required to develop additional infrastructure, some major GCC producers may curtail exports through the SOH in the interim. Nevertheless, pipelines do offer diversification benefits, reduce the vulnerability to SOH, and could signal to customers clear intention to preserve the role of a reliable and secure supplier.

[40] Reuters. 'Middle East war damage to energy assets may cost up to 58 billion, research firm Rystad says', 15 April 2026.
[41] Reuters. 'Strait of Hormuz is shut, must reopen without conditions, UAE oil giant ADNOC's CEO says', 9 April 2026.

5.4. Russia: Change of fortunes

The US-Israel war on Iran had many unintended consequences most notably the strengthening of Russia's financial position, a paradoxical outcome in the light of extensive sanctions imposed on the country's oil sector.

The imposition of US tariffs on India's Russian crude oil imports alongside US sanctions on Rosneft and Lukoil towards the end of 2025 forced India to reduce its Russian oil purchases. Russian crude exports to India averaged 1.3 mb/d between December 2025 and February 2026, down by around 400 kb/d compared with the 2025 average of 1.7 mb/d. This eroded Russian pricing power and stretched its trade logistics with Russian crude on water rising to very high levels.

One of the consequences of the war has been Russia's regaining of its pricing power, with its crude trading at a premium, boosting Russia's earnings from oil exports to the highest levels since the start of the Ukraine war.[42] India has been the biggest beneficiary, increasing its crude imports from Russia by 15% m/m in March, rebounding near 1.5 mb/d. Additionally, by the US granting waivers on sanctioned Russian oil on water in March (extended again in April), it enabled Russia to diversify its customer base, alleviating the logistical challenges. Also, with the loss of Venezuelan and potentially Iranian barrels, the strategic importance of Russian oil for China has grown. Russia's stronger fiscal position may allow it to sustain the war against Ukraine for a longer period. Ukraine in response has increased its attacks on Russian energy infrastructure to limit Russia's gains. Russian exports have barely increased due to attacks on energy infrastructure[43], amplifying the current stresses in the oil market.

5.5. China: Strong buffers but not immune to a prolonged shock

China's exposure to oil flows through the SOH is high, importing almost 50% of its crude, 40% of its naphtha and 45% of its LPG through the Strait. However, China is well positioned to mitigate the impact of the supply shock due to its strategic buffers.[44] China has been increasing its strategic storage capacity and over the last few months has accumulated large commercial and strategic stocks (**Figure 20**). According to Kpler data, China accumulated a total of 132.5 million barrels between March 2025 and March 2026. Independent refineries in Shandong have also benefited from discounted sanctioned barrels. Independent refineries still have access to these sanctioned barrels, particularly Iranian barrels as large volumes are being hoarded on tankers anchored outside the SOH,[45] although the discounts on offer have since narrowed or disappeared. The repricing of sanctioned crudes alongside controlled retail prices and restriction on exports of products have increased the pressure on their refining margins. This is testing their ability to follow government guidance to keep gasoline and diesel output at least at the 2025 level. Beijing has issued additional crude import quotas to support purchases, suggesting also that if runs drop, the independents risk losing future import quotas.

China's response to the shock has been evolving. Although it is one of the few countries with surplus refining capacity, it has banned exports of products to bolster its domestic energy security. At the start of the war, the Chinese authorities refused requests from state-owned refineries to release stocks from commercial or the strategic reserves. However, more than one month into the crisis, China is reportedly set to allow state-owned refineries to utilise commercial crude inventories. That said, little to no stocks have been drawn down[46].

[42] Bloomberg. 'Russia Boosts Oil Income to Highest Since Early in Ukraine War', 8 April 2026.

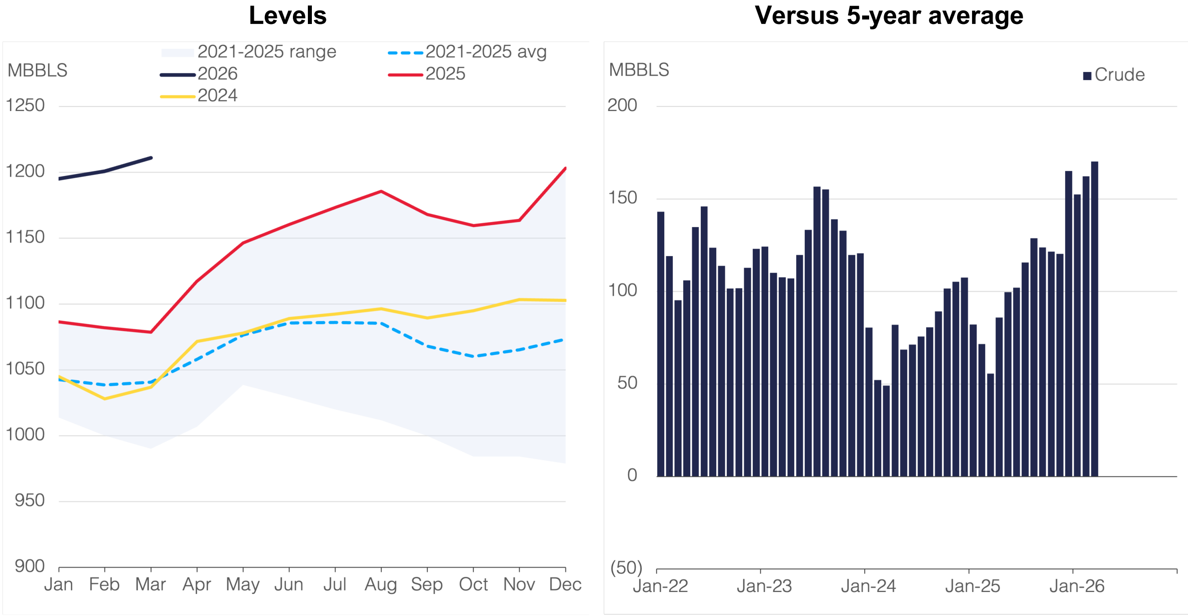
[43] Bloomberg, 'Russia Struggles to Boost Oil Flows After Black Sea Port Hobbled', 14 April 2026.

[44] Meidan, M. 2026. 'Disruption in the Strait of Hormuz: Implications for China's energy markets and policies', OIES, March.

[45] Bloomberg, 'Iran Oil Hoard at Sea Shields China's Refiners From US Blockade', 14 April 2026.

[46] Bloomberg. 'China's Oil Stockpiles Remain Robust as Teapots Crank Up Output', 24 April 2026.

Fig. 20: China's commercial and strategic crude oil stocks



Source: Kpler, OIES

Though these measures have helped China to mitigate the impacts of the initial shock, a longer disruption is likely to cause a gap in domestic crude supplies. China's crude imports in March have declined due to lower imports from the Gulf.[47] For instance, imports from Saudi Arabia, a key supplier to China, are set to halve in May from 40 to 20 million barrels.[48] China's refineries have cut runs with throughputs expected to fall by around 1.4 mb/d between February and April 2026, as they face lower availability and higher cost of crude and pressures on refining margins. Also, China remains exposed to rising prices and the compounding pressures of higher oil costs, petrochemical feedstock constraints, alongside a worsened global economic outlook. Petrochemical producers in China have been cutting operations as they face higher feedback cost and lower export demand.[49] The latest data suggest that the surge in energy costs has ended China's three-year factory deflation.[50] Therefore, while Beijing has some buffers, it is by no means sanguine or immune to the shock.

A key question remains as to the extent that China would exert its leverage with Iran to open the SOH. China's diplomacy has been largely rhetorical, enhancing its 'strategic coordination' with Pakistan to mediate between Iran and the US, and putting forward a broad proposal which calls for an immediate ceasefire, halt attacks on energy infrastructure and re-open the SOH through negotiations. China opposed a United Nations Security Council Resolution put forward by Bahrain calling for countries to use "all necessary means" to keep the SOH open. China recognises 'Iran's sovereignty security and legitimate rights as a country along the Strait of Hormuz, which explains China's preferred approach to negotiate with Iran safe passage for its vessels. Of the few oil and products tankers that departed the SOH since the start of the war, many of these are under Chinese ownership. But with the US blockade, this strategy faces a serious challenge, with some observers arguing that a key reason behind the US blockade is to push China to exert its leverage over Iran though the extent of China's influence over

[47] Bloomberg. 'China's Oil and Gas Imports Shrink on Persian Gulf Turmoil', 14 April 2026.
[48] Bloomberg. 'Saudi Oil Sales to China to Halve as Hormuz Crisis Lifts Prices', 13 April 2026.
[49] Bloomberg. 'China's Petrochemical Producers Idle Capacity as Margins Crumble', 15 April 2026.
[50] Bloomberg. 'China Ends Over Three Years of Factory Deflation After Oil Shock', 10 April 2026.

Iran remains unclear.[51] Since the US blockade, China has been more vocal, calling on all parties to guarantee the freedom and safety of navigation.

5.6. Asia: In the eye of the 'shock'

Asia has borne the brunt of the shock losing access to both crude and products supplies from the Gulf. For many countries, Gulf medium sour crudes constitute the baseload for their refineries. In addition, Asian countries rely heavily on products flows through SOH, particularly LPG and naphtha. For instance, for a country like India, more than 95% of its LPG imports pass through the SOH.

In response to the disruption, Asian refineries have been trying to mitigate the loss of Gulf crudes and products by sourcing barrels from alternative regions including the US, Latin America, and West Africa. Asian refineries have become increasingly dependent on US crude imports, mainly WTI, which has become an integral part of Asia's crude slate. The substitution of medium and heavy sour crude with light sweet crude however is not optimal as Asian refineries are configured to process medium and heavy sour crudes while lighter crudes tend to yield higher proportions of naphtha and less straight run fuel oil and VGO to feed secondary units reducing efficiency and products' output.[52]

A US waiver on sanctioned Russian and Iranian barrels provided some additional relief. For instance, India resumed its purchases of Russian oil with 60 million barrels to be delivered in April[53] and for the first time in years, it procured Iranian shipments[54]. Also, several Asian countries negotiated passage arrangements with Iran, though the volumes that passed through SOH remained modest. In parallel, OECD Asia drew on the SPR to provide additional supplies, but their impact was short-lived and some countries such as Japan has announced that it would release additional volumes.

Asia has also benefited from producers' strategic storage facilities located close to consumers.[55] Just before the war in February, some key Gulf producers hiked their production to boost stocks in these storage facilities and in tankers allowing them to supply more than what they produced in March.

Governments have also introduced support measures to cushion consumers from high prices and discourage exports of refined petroleum products (for instance, India imposed a tax on the exports of refined products).

Despite these efforts, Asia has been unable to fully replace the lost barrels from the Gulf. Moreover, securing crude and products from other regions has increased costs, amplified by elevated freight rates and greater competition and not having access to the optimal type of crude. Consequently, Asia had to balance with refineries cutting runs and consumers rationing demand. This is key as Asian demand accounts for most of the growth in global oil demand.

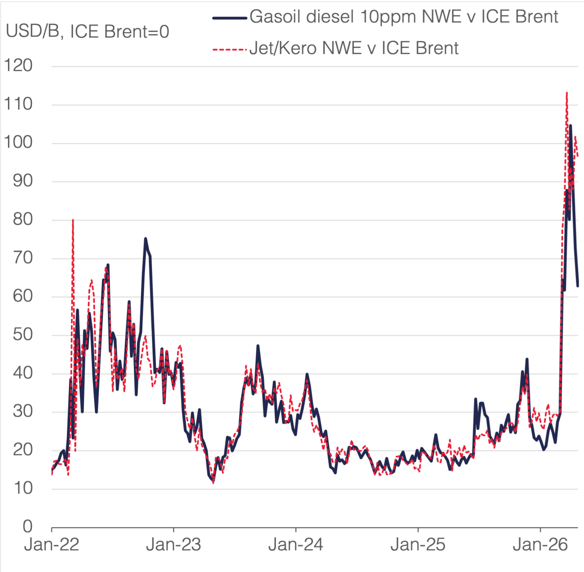
5.7. Europe: Increased vulnerability with limited influence

Although Europe's reliance on oil trade flows through the SOH is relatively limited compared to Asia, Europe can't shield itself from the broader consequences of the war[56]. In addition to being exposed to higher energy prices, European supplies of crude and products could be impacted as it faces tougher competition for the available pool of crude and products that could have been used to offset the losses from the Gulf. Premiums on medium sour crudes such as Norway's Johan Sverdrup reached record levels in Europe. The biggest impact on Europe however will be felt on products, particularly distillates and jet fuel, given its high reliance on imports from the GCC and as competition for products from other regions intensifies (**Figure 21**). Also, if EU refineries must procure crude at higher prices and settle for

[51] Forbes. 'China's Growing Interest In Opening The Strait of Hormuz', 22 April 2026.
[52] Argus. Global Markets, Volumes LVI, 15, 17 April 2026.
[53] Oil Price. 'India Snaps Up 60 Million Barrels of Russian Crude for April Delivery', 25 March 2026.
[54] Bloomberg. 'Iranian Oil Arrives in India Just Ahead of US Waiver Expiry, 15 April 2026.
[55] For instance, Aramco has such facilities in Japan, South Korea and Rotterdam.
[56] Fattouh, B. and Economou, A. 2026.'Europe's Oil Vulnerability to the Strait of Hormuz Disruption', OIES Comment, April.

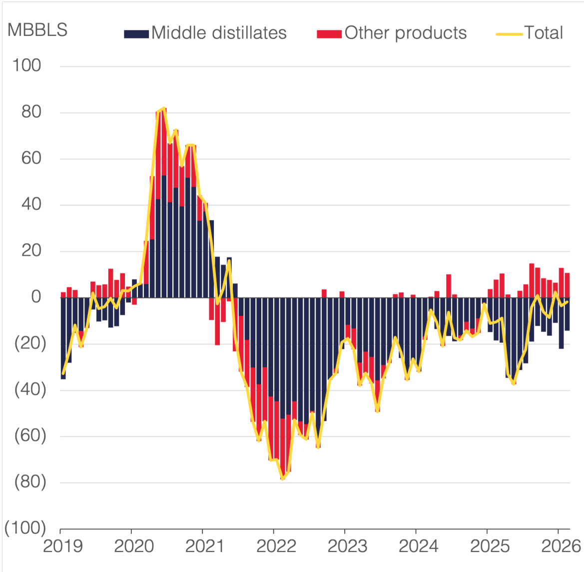
grades that don't optimise their operations and yields, then this could impact refining margins forcing them to cut runs.

Fig. 21: NWE product cracks



Notes: Data shown are weekly, as of week-ending 24 April 2026. Source: Argus, OIES

Fig 22: OECD Europe commercial stocks vs 5-year average



Source: IEA, OIES

EU's buffers such as strategic and commercial stocks are limited in the face of such a severe supply shock (**Figure 22**), and if the disruption is prolonged, those buffers will be further depleted impacting consumers and forcing EU governments to take some tough and costly measures. The European Commission has asked member countries to consider 'voluntary demand saving measures ... with particular attention to the transport sector' and 'refrain from taking measures that may increase fuel consumption, limit the free flow of petroleum products or disincentivize EU refinery output'.

At the more strategic level, despite US President Trump's pressure on European nations to help secure the SOH and while the EU reaffirms 'the importance of safeguarding maritime routes, and safety of navigation, including in the Strait of Hormuz and all associated critical waterways' and 'the absolute necessity to permanently restore safe and toll free freedom of navigation in the Strait of Hormuz, the EU leaders have so far rejected military involvement and the focus is to reopen the SOH via coalitions and diplomacy. Given its current posture, the EU is likely to have limited influence on SOH operations during the war and in shaping the post-war governance framework that emerges around SOH.

6. Conclusion

During the last decade, the oil market has been subject to a series of different types of shocks including attacks on key energy infrastructure, the COVID demand shock, disruption to oil trade flows, and the imposition of sanctions on key producing countries. Despite their severity, the oil market demonstrated remarkable resilience in absorbing these shocks: oil prices continued to trade within a narrow range and exhibited low volatility, oil supplies were plentiful, and oil continued to flow. To be clear, these shocks had profound impacts on oil markets: trade logistics became more complex and costly, trade routes became longer and had to be reconfigured, markets have become more segmented, and importers and exporters had to diversify their client base. But overall, the system absorbed these shocks and continued to function in an efficient manner. The scale of the industry, the massive infrastructure in place due to decades of investment, and the interconnectedness and maturity of markets all contributed to this resiliency.

The current supply shock is testing the key foundations of this resiliency. Flows have been disrupted with limited opportunities to divert these flows; spare capacity, the key buffer in the oil system was unavailable; key energy infrastructure has been attacked and damaged; oil trade logistics have become more complex and costly; countries have restricted exports of refined products; and the US and Iran have shown willingness to weaponize the SOH to enhance their negotiating positions. The duration of the disruption and whether a secure and free passage through SOH will be restored also remain key uncertainties.

These factors are already shaping the expectations of market participants and causing shifts in behaviours, strategies and policies of key players including geopolitical and trade relations, energy security and transition policies, stock holding policy including the proportion of oil versus products and the various crude grades, diversification of trade routes, development of new infrastructure, commercial and sales arrangements, and investment strategies both inside and outside the oil sector. Such shifts will determine oil market outcomes and whether the market will prove resilient in the longer term.